

Task 5: Dynamic Quoting Under Inventory Pressure

1. Approach and Functional Form

Our dynamic quoting strategy is designed to balance the inherent trade-off between maximizing spread capture and minimizing adverse selection and end-of-day inventory penalties. The functional form is constructed as a base spread adjusted by an inventory skew, fortified with a hard “Toxicity Shield” to reject extreme tail risks.

The Core Quoting Function:

For standard flow, the bid and ask half-spreads (δ_b, δ_a) are defined as:

$$\begin{aligned}\delta_{base} &= \sigma \cdot (c_{base} + c_\alpha \cdot \alpha) \\ \text{Skew} &= c_{inv} \cdot I \cdot e^{c_{time} \cdot \eta} \\ \delta_b &= \max(c_{min}\sigma, \min(\delta_{max}, \delta_{base} + \text{Skew})) \\ \delta_a &= \max(c_{min}\sigma, \min(\delta_{max}, \delta_{base} - \text{Skew}))\end{aligned}$$

Toxicity Shield:

If an incoming trade exhibits extreme adverse characteristics—specifically, if it originates from highly toxic clients, exceeds the 95th percentile of historical volume (> 203), or triggers a severe Machine Learning adversity score ($\alpha > 0.9$)—the algorithm instantly defends itself by quoting the maximum allowable spread ($\delta_{max} = 50$ bps). This intentionally tanks the fill probability, effectively refusing the toxic flow.

2. Intuition Behind Parameterization

The parameters were tuned via grid search on the validation set, driven by the following market microstructure intuition:

- **$c_{base} = 0.75$:** The baseline risk-premium demanded for absorbing standard inventory variance relative to local volatility.
- **$c_\alpha = 0.05$:** A targeted penalty multiplier for adverse selection. As the probability of informed trading (α) rises, the base spread widens to protect against anticipated short-term momentum drifts.
- **$c_{inv} = 0.90$:** The linear sensitivity to accumulated inventory (I). If the Liquidity Provider is long ($I > 0$), the Skew is positive. This forcibly widens the bid (discouraging further buys) and tightens the ask (encouraging sells to flatten the book).
- **$c_{time} = 3.0$:** The time-decay urgency factor. The end-of-day penalty ($\phi \cdot I^2 \cdot \sigma$) heavily punishes holding inventory at $\eta = 1$. By applying an exponential multiplier $e^{3\eta}$ to the Skew, the strategy shifts from passive spread-capture in the morning ($\eta \approx 0$) to aggressive, urgent inventory liquidation in the final hours ($\eta \rightarrow 1$).

3. Validation Methodology and Results

Because the exact fill parameters (λ, γ) and the penalty coefficient (ϕ) are concealed and subject to continuous, abrupt regime shifts, the strategy was rigorously designed to avoid overfitting to a single market state.

Methodology:

The strategy was validated using a Monte Carlo backtesting engine simulating 20 vastly diverse market

regimes (varying baseline fill rates, fill decay steepness, and penalty harshness). To eliminate random noise generated by the stochastic fill probability function ($P_{fill} = \lambda e^{-\gamma\delta/\sigma}$), each of the 20 regimes was simulated and evaluated across a 10-seed average.

Results & Robustness:

The exponential time-decay mechanism successfully flattened end-of-day inventory across all simulated regimes, rendering the unknown ϕ penalty practically negligible. Furthermore, the Toxicity Shield successfully filtered out severe drawdowns during adverse regime shifts. By relying purely on the observable state vectors $(I, \sigma, \alpha, \eta)$ rather than hardcoded global assumptions, the algorithm maintained a consistently high Sharpe-like Score, robust positive net PnL, and strictly constrained maximum drawdowns across all tested adversarial environments.